ROUTE INVESTIGATION REPORT FOR UBIE FLOW STATION TO ZARAMA NAG MANIFOLD

INTRODUCTION:

Ubie NAG development is part of Gbaran NODE opportunity aim at maturing 1260Bcf of gas to increase gas delivery to NLNG and fulfill gas supply commitment of SPDC to NLNG thereby mitigating potential delay and shortfalls in gas supply to NLNG.

OBJECTIVE:

To identify and propose gas-line route options for Gas evacuation starting from Ubie Flow Station tie-in location (**455,022.21mE, 123,179.82mN – Mid belt**) to Zarama NAG manifold site.

SUMMARY:

From the Route Assessment Statistics, Proposed Route Option 2 has the shortest route length of 27.6km while Proposed Route Option 1 and 2 have longer route length of 30.5km and 27.8km respectively.

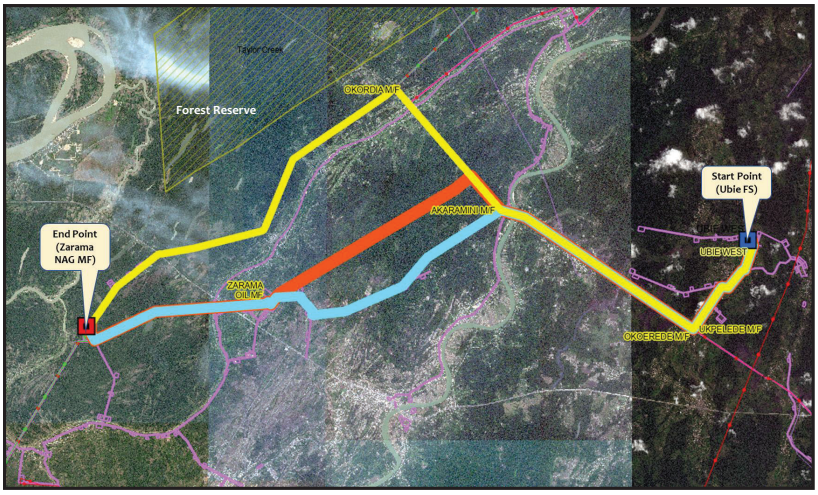
METHODOLOGY:

- Analysis of the most recent high-res satellite imagery for land-use and land cover.
- Considerations for planning the route are environmental, technical, economic and socio-cultural issues using satellite imagery and vector data.
- Generate statistical assessment for the proposed gas line route.

DATASETS USED:

Raster- High Resolution Satellite Imagery of 0.5m resolution acquired between December 2021 and January 2023.

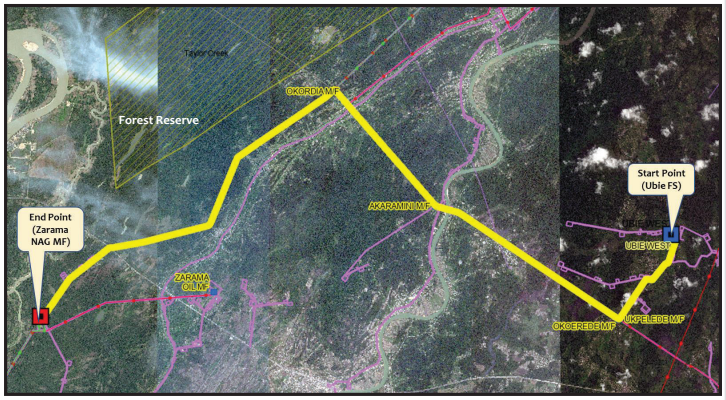
Vector- The dataset used from Geomatics corporate database are: SPDC Land Parcels Portfolio; Settlement layer data; Electric Powerlines data; Pipelines(SCiN); Flowlines(SCiN); Telecoms All; Gas Plants; Flow Station and Manifold layers.



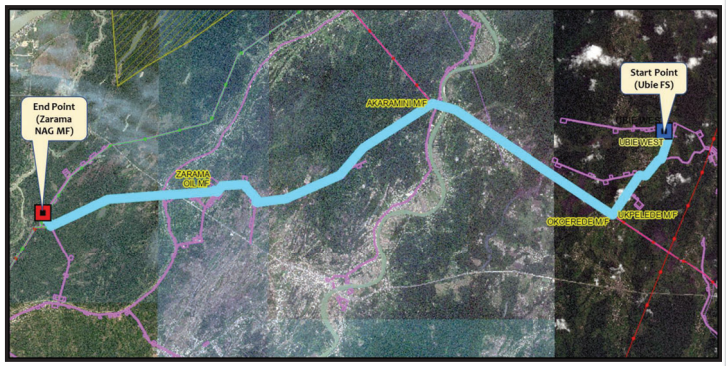
Clip1: Overview image Showing the Proposed Route Options identified for Ubie NAG development

S/N	DESCRIPTION	PROPOSED ROUTE 1	PROPOSED ROUTE 2	PROPOSED ROUTE 3
1	Approx. total length (m)	30.5	27.6	27.8
2	Length through existing ROW (m)	30.5	24.8	20.2
3	Length through new ROW m)	Nil	2.8	7.6
4	Proportion of existing ROW used (%)	100%	89.9%	72.7%
5	Proportion through new ROW (%)	Nil	10.1%	27.3%
6	No of Communities within 2km buffer	**21** (AYAMABELE, NEW-JERUSALEM, IDU-EPKEYE, EBRASS, ZARAMA, UKPELIEDE, AKARAMIRIN, AKALA-OLU, AKUMONI, IKARAMA, EBERIABA, KALABA, ZARAMA-NYAMBIRI, OKOTIAMA, JOINKRAMA, OSHI, ZARAMA, KARAMA, IDU, AKUMANI, OBAREKKAI)	**13** (NEW-JERUSALEM, AGBOBIRI, IDU-EPKEYE, EBRASS, ZARAMA, OSHE, UKPELIEDE, AKARAMIRIN, AKALA-OLU, EBERIABA, OKOTIAMA, OSHI, IDU)	**16** (NEW-JERUSALEM, AGBOBIRI, IDU-EPKEYE, EBRASS, ZARAMA, UKPELIEDE, AKARAMIRIN, AKALA-OLU, EBERIABA, ZARAMA-NYAMBIRI, OKOTIAMA, JOINKRAMA, OSHI, IDU, AKUMANI, OBAREKKAI)
7	No. of Forest reserve traversed	Nil	Nil	Nil
8	Pipeline/Flowline crossings	4	2	2
9	No of Telecom(Nitel) Cable crossings	Nil	Nil	Nil
10	No of Electrical Cable crossings	Nil	Nil	Nil
11	Length of Secondary/Sparse Vegetation Forest traversed (m)	30.5	27.6	27.8
12	Length of Mature Forest traversed (m)	Nil	Nil	Nil
13	Length of Swamp Vegetation traversed (m)	Nil	Nil	Nil
14	Length of Built Up traversed (km)	5.7	5.2	5.7
15	River Crossing	2	3	3
16	River Width (m)	8.84, 139.12	8.84, 139.13, 40.3	8.84, 139.13, 40.4
17	Major Road Crossings	9	7	7

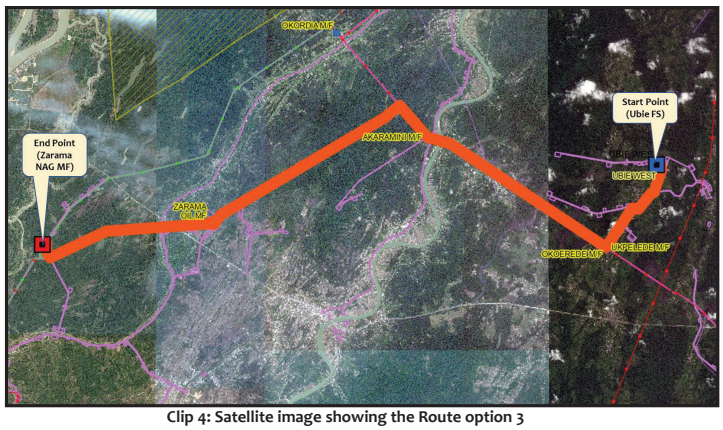
Table 1: Route Assessment Statistics



Clip 2: Satellite image showing the Route Option 1



Clip 3: Satellite image showing the Route Option 2



Clip 4: Satellite image showing the Route option 3

CONCLUSION:

From the route assessment statistics, three proposed route options were proffered and investigated for this project. As required, several vital statistics based on the Environmental and Safety, Engineering and Operational, Biodiversity considerations were analyzed and extracted in the Route Assessment Statistical table above. The route options were not ranked in this study but left for ranking during the multi discipline route selection workshop using the SCAN tool.